Sparse-View RGB-D 3D Reconstruction

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Team Setup

Student information

- Nguyễn Ngọc Minh – 20235533
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Presentation goal

Present the final RGB-D/source-depth reconstruction setting and the Run 30 source-ray correction result after the RGB-only learned extensions failed reconstruction-level gates.

Motivation

- 3D reconstruction is important because 3D representations help machines understand spatial structure, depth, and object relationships more accurately.
- Practical 3D capture often has only a small number of usable images.
- Classical high-quality pipelines depend on accurate camera poses or expensive scene-specific optimization.
- Sparse views increase ambiguity because cross-view correspondence is harder to establish reliably.
- A fast, generalizable method for unseen scenes would be useful for robotics, AR/VR, navigation, and scalable environment digitization.

Problem Statement

Core question

How can we reconstruct unseen indoor scenes accurately when only 2 to 5 sparse posed RGB-D views are available?

Key constraints

- Unseen scene at test time
- Sparse posed RGB-D views
- Known camera intrinsics/extrinsics
- No per-scene optimization

Expected outcome

- Robust geometry recovery
- Stable performance under view reduction
- Better occlusion and wrong-depth behavior
- Practical runtime per scene

Objectives

1. Build a strong sparse-view MV-DUSt3R+ baseline for 2–5 views.
2. Diagnose why RGB-only learned reliability, match filtering, and monodepth correction fail reconstruction-level gates.
3. Switch the final setting to posed RGB-D views with known camera intrinsics/extrinsics.
4. Validate Run 30 source-depth correction on overall, occlusion, and repeated/wrong-depth ambiguity subsets.

Final Hypothesis

Working hypothesis

Candidate deletion is too risky for sparse-view reconstruction. When input source depth is available, anchoring source rays to metric RGB-D geometry should correct wrong-depth candidates while preserving recall.

- Keep the useful MV-DUSt3R+ candidate set instead of over-filtering it
- Use source depth, poses, and intrinsics to correct 3D locations
- Test the result on occlusion and repeated/wrong-depth hard subsets

Proposed Contributions

Our contribution

The final contribution is a sparse-view RGB-D source-depth correction pipeline. RGB-only learned modules are kept as diagnostic evidence, while Run 30 is the accepted method.

- **Strong RGB-only baseline:** Run 11 improves B0 across 2–5 views using view selection and fixed confidence thresholds.
- **Negative RGB-only analysis:** Runs 22–29 show that proxy F1, candidate filtering, and generic monodepth do not solve reconstruction.
- **Final RGB-D method:** Run 30 uses input source depth maps with poses/intrinsics for source-ray correction and passes all gates.

Contribution Details

1. Baseline that solves sparse-view instability

- Run 11 fixed thresholds and view selection beat B0 on 2, 3, 4, and 5 views.
- This solves the first limit but leaves occlusion and wrong-depth ambiguity as hard cases.

2. RGB-only diagnostic path

- OARH, RSDH, RAJAH, and monodepth corrections are tested through reconstruction-level gates.
- They expose why filtering hurts recall/completeness and why generic monodepth is not enough.

3. Run 30 source-depth correction

- Uses source RGB-D depth maps with known camera poses/intrinsics.
- Corrects source rays in metric geometry instead of deleting candidates.

Why RGB-only Failed

Transition narrative

Runs 22–29 showed that RGB-only filtering/correction was insufficient. Run 30 changes the inference contract to RGB-D and passes the overall, occlusion, and ambiguity gates.

- Proxy F1 did not transfer to reconstruction-level F-score.
- Filtering removed valid sparse-view geometry and hurt recall.
- All-candidate retention became a strong non-learned baseline.
- RGB-only monodepth did not recover metric/source depth well enough.

Proposed Method

- Primary method: **MV-DUSt3R+**
- Single-stage feed-forward multi-view reconstruction
- Final setting uses sparse posed RGB-D views
- Run 30 adds source-depth correction after MV-DUSt3R+
- Our study focuses on robustness, hard cases, and runtime

Source: Tang et al., MV-DUSt3R+, project page and paper

Architecture Comparison

DUSt3R baseline

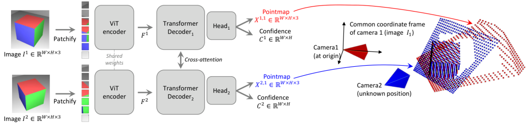
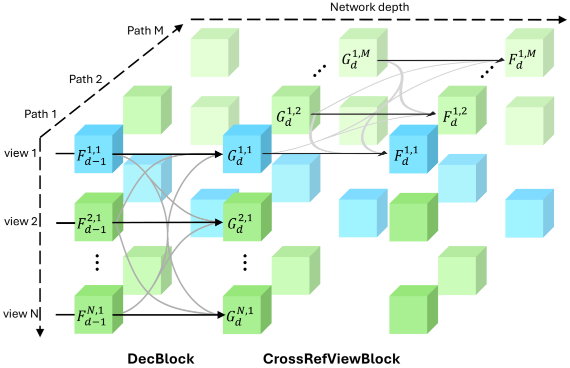


Figure 2. **Architecture of the network.** Two views of a scene (I^1, I^2) are first encoded in a Siamese manner with a shared ViT encoder. The resulting token representations F^1 and F^2 are then passed to two transformer decoders that constantly exchange information via cross-attention. Finally, two regression heads output the two corresponding pointmaps and associated confidence maps. Importantly, the two pointmaps are expressed in the same coordinate frame of the first image I^1. The network is trained using a simple regression loss (Eq. (4))

Pairwise reasoning with cross-attention between two views

MV-DUSt3R+ main method



Multi-path design with cross-reference-view information flow

Source: Left: Wang et al., DUSt3R, CVPR 2024. Right: Tang et al., MV-DUSt3R+, project page/paper

Why MV-DUSt3R+ vs DUSt3R?

Aspect	DUSt3R	MV-DUSt3R+
Processing style	Pairwise reconstruction	Joint multi-view reconstruction
Global consistency	Requires later alignment	Built into one forward pass
Sparse-view behavior	Can accumulate pairwise errors	More robust cue fusion
Runtime trend	Slower as views increase	Better scaling with more views
Role in project	Reference baseline	Main candidate method

- DUSt3R is still a strong baseline, so the comparison is direct and meaningful.
- MV-DUSt3R+ is the candidate generator for our final RGB-D correction step.

Quantitative Evidence from the Paper

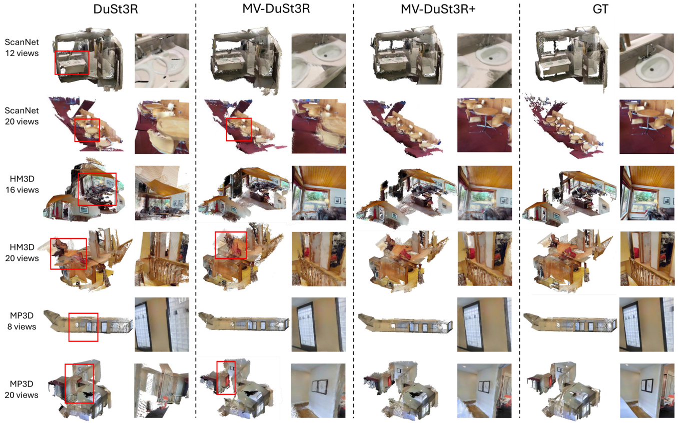
	Method	GO	HM3D ND ↓	HM3D DAc ↑	HM3D CD ↓	ScanNet ND ↓	ScanNet DAc ↑	ScanNet CD ↓	MP3D ND ↓	MP3D DAc ↑	MP3D CD ↓	Time (sec)
4 views	Spann3R	×	37.1	0.0	225(184)	8.9	19.5	54.7(50.1)	42.7	0.0	248(202)	0.36
	DUSt3R	✓	1.9	75.1	5.6(2.3)	1.3	89.8	4.0(0.4)	3.9	41.7	40.0(5.3)	2.42
	MV-DUSt3R	×	1.1	92.2	2.0(1.1)	1.0	93.3	2.0(0.4)	2.5	62.4	25.3(4.1)	0.05
	MV-DUSt3R+	×	1.0	95.2	1.5(0.9)	0.8	94.9	1.5(0.3)	2.2	68.0	19.9(3.4)	0.29
	MV-DUSt3R$_{oracle}$	×	1.0	94.6	1.5(0.7)	0.8	95.5	1.3(0.3)	2.3	66.6	20.7(4.0)	-
	MV-DUSt3R+$_{oracle}$	×	0.9	96.5	1.4(0.7)	0.7	95.8	1.2(0.2)	2.1	70.6	17.9(3.3)	-
12 views	Spann3R	×	32.6	0.0	125(113)	9.1	16.3	36.6(31.2)	35.0	0.0	138(112)	1.34
	DUSt3R	✓	3.9	30.7	18.1(3.4)	1.9	82.6	4.1(0.6)	6.6	12.0	49.6(8.3)	8.28
	MV-DUSt3R	×	1.6	79.5	3.0(1.2)	1.4	86.8	2.3(0.8)	3.4	41.3	22.6(5.5)	0.15
	MV-DUSt3R+	×	1.2	91.5	1.8(0.7)	1.2	88.4	1.8(0.7)	2.6	55.0	15.1(3.8)	0.89
	MV-DUSt3R$_{oracle}$	×	1.3	88.8	1.8(0.9)	1.0	90.6	1.3(0.7)	2.9	51.3	16.4(4.0)	-
	MV-DUSt3R+$_{oracle}$	×	1.1	94.8	1.4(0.7)	1.0	90.9	1.3(0.5)	2.5	59.8	13.6(3.5)	-
24 views	Spann3R	×	41.7	0.0	139(121)	11.4	1.6	37.4(35.5)	46.6	0.0	151(121)	2.73
	DUSt3R	✓	6.8	7.3	32.4(5.2)	2.4	72.6	5.1(1.0)	11.4	2.5	80.9(14.3)	27.21
	MV-DUSt3R	×	3.4	36.7	10.0(3.5)	2.2	75.2	2.7(0.9)	6.3	12.2	38.6(13.9)	0.35
	MV-DUSt3R+	×	2.1	64.5	3.9(2.0)	1.6	81.2	1.7(0.7)	4.3	26.7	22.0(5.9)	1.97
	MV-DUSt3R$_{oracle}$	×	2.1	58.9	3.5(2.1)	1.4	82.9	1.4(0.7)	4.4	22.0	19.9(5.1)	-
	MV-DUSt3R+$_{oracle}$	×	1.8	77.9	2.6(1.3)	1.3	85.1	1.3(0.6)	3.6	33.1	15.1(4.4)	-

Table 2 MVS reconstruction results. Best results are marked in red. For clarity, metric ND is scaled up by 10×, DAc is in percent %, and CD is scaled by 100× with its median given in parentheses. Results of our **oracle** methods are obtained by manually selecting the best single reference view from reference view candidates to report a possibly achievable

Across multiple datasets and view counts, the highlighted rows show that MV-DUSt3R+ improves reconstruction quality while keeping runtime practical.

Source: Tang et al., MV-DUSt3R+, Table 2

Qualitative Motivation



DUSt3R often fails on repeated structures, while MV-DUSt3R+ remains more stable in sparse-view scenes.

Source: Tang et al., MV-DUSt3R+, qualitative results

Final Pipeline

Select sparse posed RGB-D views and scene-level train / validation / test splits

↓

Run MV-DUSt3R+ to produce candidate point cloud and confidence

↓

Use source depth maps with poses/intrinsics to anchor source rays

↓

Apply Run 30 selective source-depth correction

↓

Evaluate overall, occlusion, and repeated/wrong-depth gates

Experiment Variables

Controlled factors

- Number of input views
- RGB-D source depth availability
- Scene-level train / validation / test split
- Fixed Run 30 policy on final test

Observed outcomes

- Reconstruction accuracy
- Completeness and consistency
- Runtime per scene
- Occlusion and repeated/wrong-depth failures

RGB-only Diagnostic Path

1. First build a supervised label cache with per-view visibility, occlusion, floating/wrong-depth, and match labels.
2. Mine occlusion-heavy, low-overlap, and hard-negative subsets before training the learned heads.
3. Freeze MV-DUSt3R+ and train small OARH / RSDH heads from generated labels, excluding direct target-label leakage features.
4. Runs 22–26 move the heads into reconstruction-level validation gates with fixed-confidence, top-k, and all-candidate baselines.
5. Run 27 is self-contained and learns a bounded confidence residual from actual candidates, self-geometry support, and raw image-patch cues.
6. Completed Run 27 still selects all candidates: learned filtering beats fixed confidence but loses completeness against full retention.
7. These runs motivate switching the final inference contract to RGB-D.

RGB-D Source-Depth Correction

1. Run 28 therefore preserves candidate count and learns source-ray 3D corrections from training depth and poses, using no depth at inference.
2. Completed Run 28 still fails the gate, but its source-depth oracle shows large occlusion and ambiguity headroom.
3. Run 29 approximates that oracle with RGB-only monocular depth aligned through the input poses and intrinsics.
4. Completed Run 29 regresses both occlusion and ambiguity, so RGB-only monodepth is not enough.
5. Run 30 makes source depth an explicit RGB-D input and gates full/selective source-ray correction policies.
6. Selected policy: `rgbd_residual_ge_0.30`, residual mode, $\alpha = 1.0$, threshold $0.30m$.
7. Final selected method: `rgbd_source_depth_selected`.
8. Keep final test clean: no threshold tuning, no checkpoint selection on test scenes.

Evaluation Data Used

Controlled ScanNet-style RGB-D subset

The reported Runs 19–31 use posed indoor RGB-D scenes with source images, depth maps, and known camera intrinsics/extrinsics.

Scene-level split

- 18 train scenes
- 6 validation scenes
- 6 held-out test scenes
- No test scene used for training or policy selection

Sparse-view protocol

- 3, 4, and 5 selected views
- Hybrid and diversity-aware view groups
- Geometry metrics at a 0.05 m threshold
- Hard occlusion and ambiguity subsets

Evaluation Scope

What the current evidence supports

Held-out comparison of Run 30 against all-candidate and fixed-confidence baselines on the controlled ScanNet-style subset.

What is not claimed

The current results are not a full ScanNet++ evaluation and do not establish official-benchmark generality. A larger mesh or laser-scan benchmark remains future work.

- Run 11 remains the strongest RGB-only baseline.
- Runs 12–29 are diagnostic and negative evidence.
- Run 30 is the final RGB-D/source-depth method.

Evaluation Plan

- Reconstruction F-score against available ground-truth geometry
- Accuracy, completeness, and Chamfer distance of point clouds
- Sparse 2–5 view behavior across scene splits
- Limit-specific gates for occlusion and wrong-depth ambiguity
- Runtime per scene for practical inference

	Method	GO	HM3D			ScanNet			MP3D			Time
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Source: Tang et al., MV-DUSt3R+, quantitative benchmark

Success criterion

Run 30 must beat the best non-learned baseline overall and improve both occlusion and ambiguity subsets by at least the gate margin.

Final RGB-D Result

Run 30 gate decision

`selected_method = rgbd_source_depth_selected`; best baseline = `all_candidates`; gate margin = 0.005; `pass_all_limits = 1`.

Split/subset	All cand.	Fixed conf.	RGB-D selected	Delta vs all
Val overall	0.1194	0.1123	0.1753	+0.0559
Val occlusion	0.1064	0.0917	0.2522	+0.1458
Val ambiguity	0.1623	0.1416	0.3000	+0.1377
Test overall	0.1758	0.1662	0.2764	+0.1007
Test occlusion	0.2111	0.1866	0.3146	+0.1035
Test ambiguity	0.1621	0.1462	0.2948	+0.1327

Why RGB-D Works

Final wording

RGB-only learned extensions did not pass reconstruction-level gates. After switching the inference contract to RGB-D, Run 30 uses input source depth maps with known camera poses/intrinsics for source-ray correction and passes the overall, occlusion, and ambiguity gates on held-out scenes.

- Source depth anchors each source ray to metric geometry.
- Correction preserves candidate count, unlike filtering that hurts recall and completeness.
- Occlusion and repeated/wrong-depth cases improve because the method fixes geometry rather than discarding sparse-view evidence.

Run 31 Coverage Stress Test

Purpose

Validate the frozen Run 30 method over more sparse-view groups. Run 31 does not train, tune, or select a new reconstruction method.

Coverage

- 30 scenes
- 12 groups per scene
- 360 groups total
- 3, 4, and 5 views
- No dense-frame inference

Fixed evaluation

- Hybrid and diversity-aware groups
- Two frame variants per configuration
- Frozen `residual_ge_0.30` policy
- Paired F-score delta vs all candidates
- Scene-cluster bootstrap 95% CI

Current status

Submitted/pending result. No additional stability claim is made yet.

Direct RGB-D Backprojection Baseline

Run 32 question

How well does direct depth backprojection reconstruct the same sparse-view groups without MV-DUSt3R+?

- Back-project valid depth with known intrinsics/poses and attach RGB.
- Apply fixed 2 cm voxel downsampling and cap at 3,500 points.
- Reuse Run 30 groups, metrics, and hard subsets; no policy tuning.

Subset	**Direct voxel**	**Run 30**	**Direct sampled**
Val overall	0.0874	0.1753	0.8500
Test overall	0.1359	0.2764	0.8666

Interpretation

Direct backprojection is not source-depth correction. The primary voxel direct baseline loses to Run 30, including occlusion and ambiguity subsets. The sampled diagnostic is very high because GT uses the same selected input depth maps; treat it as evaluator-circularity evidence, not an official mesh/laser-scan benchmark result.

Final Deliverables

- End-to-end implementation for Kaggle training/evaluation diagnostics
- Quantitative results on unseen scenes across multiple view counts
- Qualitative 3D visualizations and failure-case analysis
- Final report discussing RGB-D source-depth strengths and limitations
- Reproducible Kaggle experiment scripts organized under `scripts/kaggle`
- Final validation rerun script with a P100-compatible Torch fallback for unstable Kaggle GPU allocation
- Submitted supervised-extension kernels for Run 12–18 and Phase 3 label/subset kernels for Runs 19–20
- Submitted OARH v2 reconstruction path for Runs 21–23, exposing proxy-to-geometry transfer limits
- Submitted RSDH v2 reconstruction path for Runs 24–26, where proxy match validity fails the geometry gate
- Redesigned Run 27 as a reconstruction-aware ensemble; its gate selects all-candidate retention over learned filtering
- Added Run 28 source-ray correction to repair wrong depth without deleting valid occluded geometry
- Added Run 29 monodepth source-ray correction to approximate the Run 28 oracle without GT depth at inference
- Added Run 30 RGB-D source-depth correction as the final accepted solution for the two remaining limits
- Added Run 31 as a frozen-policy coverage stress test over 360 sparse-view groups; results are pending
- Added Run 32 as a direct RGB-D backprojection baseline without MV-DUSt3R+; primary voxel baseline loses to Run 30, while sampled direct output exposes evaluator circularity
- Clean GitHub repository with curated docs, proposal sources, and PDF deck

Final Contributions

- A strong RGB-only baseline that solves sparse-view instability through view selection and fixed confidence thresholds.
- A negative-analysis record showing why OARH/RSDH/RAJAH and monodepth should not be claimed as RGB-only solutions.
- A final RGB-D/source-depth method that passes reconstruction-level gates for overall quality, occlusion, and repeated/wrong-depth ambiguity.
- A clear limitation boundary: RGB-only is baseline/diagnostic; Run 30 is the final source-depth contribution.

References

- Wang et al., "DUSt3R: Geometric 3D Vision Made Easy," CVPR 2024.
- Tang et al., "MV-DUSt3R+: Single-Stage Scene Reconstruction from Sparse Views In 2 Seconds," CVPR 2025.
- Yeshwanth et al., "ScanNet++: A High-Fidelity Dataset of 3D Indoor Scenes," ICCV 2023.
- DTU Robot Image Data Sets, Multi-View Stereo Benchmark.